

Individual Reflection: Lab 5 (AutoML)

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Course: AIG130 – Cloud Computing for Machine Learning

1. The "Black Box" Problem: Explainability in AutoML

During this lab, I used AWS AutoGluon to predict Titanic survivors. While the results were impressive (about 85% accuracy), I realized that the implementation of AutoML is essentially a "black box." In our Group Report, we noted that the tool automatically trained 140 different models, including CatBoost and Neural Networks.

However, as a student, I found it difficult to understand *why* the model made certain decisions. Because AutoML hides the complex math and logic behind a simple interface, it lacks "explainability." If a model predicts that a specific passenger wouldn't survive, I can't easily see which feature was the most important or how the weights were balanced. This makes it hard to trust the model completely in real-world situations where we need to explain our results.

2. Can AutoML Replace Traditional Learning?

In my opinion, the answer is no. While tools like AutoGluon make it much faster to build models with "very little code," they cannot replace a solid foundation in machine learning.

If I didn't already know about data preprocessing or what "accuracy" and "overfitting" meant, I wouldn't be able to tell if my AutoML results were actually good or just a mistake. Using AutoML is like using a calculator: it is a powerful tool, but you still need to understand how long division works to know if the calculator's answer makes sense. We still need to learn the basic algorithms and manual coding to be truly successful in this field.

3. The Risks of Relying on Closed-Source Cloud Products

Using closed-source products like Google AutoML or AWS services has its downsides for students. When we use these "point-and-click" websites, we are seeing the final product but not the engine inside.

Since these tools are proprietary, we cannot see the source code or the specific logic the company uses to optimize the models. This creates a "dependency" on a specific brand. If we only learn how to use one specific cloud platform, we might feel lost if we have to work on a different system later. For learners, it is much better to understand the open-source logic first so that we can adapt to any tool in the future.

4. Lessons Learned and Future Changes

Working in this lab taught me that modern tools can do in minutes what used to take days of manual work. However, the biggest lesson is that human judgment is still the most important

part of the process. We still need to decide which information (features) to include.

If I were to do this lab again, I would spend more time comparing the AutoML results with a manual model I built myself. This would help me see exactly where the "black box" is performing better and where it might be making errors that a human could catch.